MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

AMINOGLYCOSIDE DOSING AND MONITORING RECOMMENDATIONS IN PATIENTS ON PEDIATRIC SERVICE LINES

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Useful links:

Pharmacokinetics Webpage

Pharmacokinetic Review (Pediatric)

Gentamicin/Tobramycin

Patient Population/Unit/Service	Empiric Dosing Recommendations	Timing of Initial Serum Concentrations
General dosing for children, EXCEPT in the following situations: • Cystic fibrosis • NICU patients • PICU patients • CHC patients	*Normal Renal Function:* Treatment: • Infants ≤5 kg: Follow NICU dosing • <18 years and >5 kg: 7.5 mg/kg/dose IV q24h • ≥18 years: Refer to Aminoglycoside Dosing in Adult Patients Synergy (for endocarditis): 1 mg/kg/dose IV q8h *Renal Dysfunction:* Acute renal insufficiency (CrCl <30 mL/min): 5 mg/kg/dose IV q24h Peritoneal dialysis: 2.5 mg/kg/dose IV q24h Hemodialysis: 2.5 mg/kg/dose IV x1 post dialysis *Obesity:* • There are limited data available to inform aminoglycoside dosing in pediatric patients with obesity.[ref] If a patient's total body weight (TBW) exceeds their ideal body weight (IBW) by more than 30%, consider using adjusted body weight (AdjBW) for aminoglycoside dosing. ○ For patients <5 feet: IBW = (Height [cm]^2 x 1.65)/1,000 ○ For male patients >5 feet: IBW = 39 + [2.27 x (Height [in] – 60)] ○ For female patients >5 feet: IBW = 42.2 + [2.27 x (Height [in] – 60)] ○ AdjBW = [0.4 x (TBW – IBW)] + IBW	• Treatment: if continuing past 48 hours, 3-hour peak and 10-hour random concentrations after the 2nd dose • Synergy: peak and trough concentrations around the 3rd or 4th dose • Acute renal insufficiency: peak and trough concentrations around the 3rd or 4th dose • PD: check trough prior to the 2nd dose • HD: check random level 2 hours after completion of next dialysis session to guide further dosing
Cystic Fibrosis Patients *Tobramycin is the preferred aminoglycoside over gentamicin*	High dose, extended interval dosing: 10 mg/kg/dose IV q24h	• 3-hour peak and 10-hour random concentrations after the 2nd dose • If concern for renal dysfunction: check trough prior to the 2nd dose • Check doses from recent previous admissions; initiate most recent therapeutic dose; discuss with pharmacy if previous dose not available

Gentamicin/Tobramycin

Patient Population/Unit/Service	Empiric Dosing Recommendations	Timing of Initial Serum Concentrations
NICU	See tables below	• Serum concentrations generally not necessary for 36-48 hour sepsis evaluations, unless at risk for/presence of renal insufficiency • Treatment: if continuing past 48 hours, peak and trough concentrations around the 2nd or 3rd dose • ECMO: trough prior to the 2nd dose • Cooling: consider trough prior to the 2nd dose if unstable renal function or pressor requirements • PD: dose by levels

Weight/PNA	0-14 days	15-28 days	>28 days
≤1,200 g	5 mg/kg q48h	4 mg/kg q36h	4 mg/kg q24h
1,200-2,000 g	5 mg/kg q36h	4 mg/kg q24h	4 mg/kg q24h
>2,000 g	5 mg/kg q24h	4 mg/kg q24h	4 mg/kg q24h

Therapeutic Hypothermia & ECMO – obtain trough prior to 2nd dose	
≤2,000 g	5 mg/kg q48h
>2,000 g	5 mg/kg q36h

Peritoneal Dialysis – dose by levels	
All weight & PNA	2.5 mg/kg q24h

Transfer from OSH – redose based on OSH dosing	
If dose ≥3.5 mg/kg:	Place patient in appropriate dosing interval as above
If dose <3.5 mg/kg:	Contact pharmacy for appropriate dosing recommendations

Gentamicin/Tobramycin		
Patient Population/Unit/Service	**Empiric Dosing Recommendations**	**Timing of Initial Serum Concentrations**
PICU	Treatment, normal renal function: • Infants ≤5 kg: Follow NICU dosing • <18 years and >5 kg: 7.5 mg/kg/dose IV q24h • ≥18 years: Refer to Aminoglycoside Dosing in Adult Patients Synergy (for endocarditis): 1 mg/kg/dose IV q8h Acute renal insufficiency (CrCl <30 mL/min), ECMO, CRRT: 5 mg/kg/dose IV q24h Hemodialysis: 2.5 mg/kg/dose IV x1 post dialysis Peritoneal dialysis: 2.5 mg/kg/dose IV q24h	• Treatment: if anticipated duration >48h or confirmed Gram-negative infection, obtain 3-hour peak and 10-hour random concentrations after the 2nd dose • Synergy: peak and trough concentrations around the 3rd or 4th dose • Acute renal insufficiency, ECMO, CRRT, PD: obtain 3-hour peak and 10-hour random level after the 1st dose to guide further dosing • HD: check random level 2 hours after completion of next dialysis session to guide further dosing
CHC (PCTU and 11W)	Normal renal function: • Infants ≤5 kg: Follow NICU dosing • <18 years and >5 kg: 7.5 mg/kg/dose IV q24h • ≥18 years: Refer to Aminoglycoside Dosing in Adult Patients Renal dysfunction, ECMO: 5 mg/kg/dose IV q24h Peritoneal dialysis: 2.5 mg/kg/dose IV q24h Synergy (for endocarditis): • Within 72 hours of cardiac surgery and/or in PCTU on inotropes/vasopressors: 1 mg/kg/dose IV q12h • ≥72 hours since last cardiac surgery and not on inotropes/vasopressors: 1 mg/kg/dose IV q8h	• Trough concentrations generally not necessary for 48-hour rule out sepsis evaluations, unless at risk for/presence of renal insufficiency • Treatment: if anticipated duration >48h or confirmed Gram-negative infection, obtain 3-hour peak and 10-hour random concentrations after the 2nd dose • Synergy: peak and trough concentrations around the 3rd or 4th dose • If concern for renal dysfunction: check trough prior to the 2nd dose

Amikacin		
Patient Population/ Unit/Service	**Dosing Recommendations**	**Timing of Initial Serum Concentrations**
General dosing for children with normal renal function, EXCEPT in the following situations: • **Cystic fibrosis** • **NICU patients** • **CHC patients** • **ECMO**	<18 years: 20 mg/kg/dose IV q24h (normal dose range 15-30 mg/kg/dose IV q24h) ≥18 years: Refer to Aminoglycoside Dosing in Adult Patients	• If continuing past 48 hours: 3-hour peak and 10-hour random concentrations after the 2nd dose • If concern for renal dysfunction: check trough prior to the 2nd dose
Cystic Fibrosis Patients	High dose, extended interval dosing: 30 mg/kg/dose IV q24h	• High dose, extended interval: 3-hour peak and 10-hour random concentrations after the 2nd dose • If concern for renal dysfunction: check trough prior to the 2nd dose • Check doses from recent previous admissions; initiate most recent therapeutic dose; discuss with pharmacy if previous dose not available
Dosing for Renal Replacement Therapy or ECMO	ECMO: 15 mg/kg/dose IV q24h CRRT: 15 mg/kg/dose IV q24h Hemodialysis: 5 mg/kg/dose IV x1 post dialysis	• ECMO or CRRT: obtain 3-hour peak and 10-hour random level after the 1st dose to guide further dosing • HD: check random level 2 hours after completion of next dialysis session to guide further dosing

Gentamicin/Tobramycin Goals (in mcg/mL)	Traditional Dosing		Extended-Interval Dosing		
	Goal peak	Goal Trough	Goal Peak	Goal Trough	18-hr Level
Gram positive synergy	3-5 (1 mg/kg) N/A (3 mg/kg)	<1	Do not use EIDA		
Gram-negative coverage in patients ≤5 kg	N/A	N/A	8-12	<1	N/A
Non-CF Gram-negative infections in patients >5 kg	8-12	<1	20-30	Undetectable (<0.25)	<1
Cystic fibrosis	10-12	<1.5	20-40	Undetectable (<0.25)	<1

Amikacin Goals (in mcg/mL)	Traditional Dosing		Extended-Interval Dosing		
	Goal peak	Goal Trough	Goal Peak	Goal Trough	18-hr Level
Non-CF Gram-negative infections	25-35	<6	40-60	<1	<4
Cystic fibrosis	25-40 (q8h) 40-60 (q12h)	<6 <6	80-120	<1	<4

Baseline Monitoring

- Serum creatinine
- Urine output

Ongoing Lab Monitoring

- **Serum creatinine**
  - ICU: every 1-3 days
  - Floor: every 3-5 days
- **Urine output** - daily

Ongoing Drug Levels

- **Extended-Interval Dosing**
  - Not needed for most 36-48 hour rule outs unless cultures turn positive
  - If GNR + culture, obtain 3-hour peak and 10-hour random levels around 2nd or 3rd dose
  - Obtain 18-hour levels every 5-7 days once therapeutic levels achieved
- **Traditional Dosing**
  - Aminoglycoside peaks
    - Not needed for most 36-48 hour rule outs unless cultures turn positive
    - After 3rd-4th dose if GNR + culture
  - Aminoglycoside troughs
    - Not needed for most 36-48 hour rule outs, unless at risk for or presence of renal insufficiency or cultures turn positive
    - ICU: Every 3-5 days
    - Floor: Every 5-7 days
    - Heme/onc: Twice weekly

CW Decentralized Pharmacist Committee Approval:	11/2024	Originated:	02/2021
Antimicrobial Subcommittee Approval:	01/2025	Last Revised:	11/2024
P&T Approval:	10/2016, 11/2022, 01/2025		

Revision History:
07/2021: Updated PH/PBM dosing recommendations
09/2022: Updated dosing for all services to include extended interval recommendations, additional guidance included related to renal dysfunction and timing of initial serum concentrations
11/2024: Standardized NICU dosing to apply to all infants <5 kg, added guidance for dosing in obesity, linked to adult guidelines for patients ≥18 years old

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines havemade all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.
If obtained from a source other than med.umich.edu/asp, please visit the webpage for the most up-to-date document.